Summary in Graph

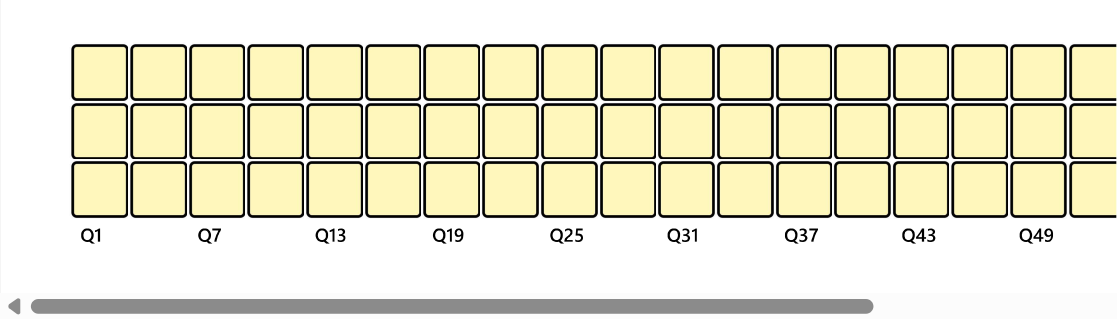
Exam Summary (GO Classes DA Test Series 2026 | MOCK GATE | Test 8)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained

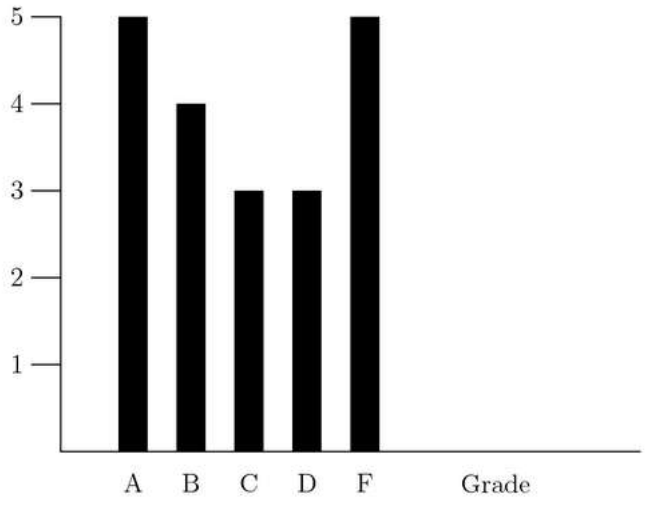


* Average Stats based on first-attempt stats of Top 10% (14) out of 140 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others



The bar graph shows the grades in a mathematics class for the last grading period. If $\mathrm{A}, \mathrm{B}, \mathrm{C}$, and D are satisfactory grades, what fraction of the grades shown in the graph are satisfactory?

A. $\frac{1}{2}$

B. $\frac{2}{3}$

C. $\frac{3}{4}$

D. $\frac{4}{5}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Choose the most appropriate word from the options given below to complete the following sentence:

If we manage to _____ our natural resources, we would leave a better planet for our children.

A. uphold
B. restrain
C. cherish
D. conserve

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 02sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Real numbers a and b satisfy the equations $3^a = 81^{b+2}$ and $125^b = 5^{a-3}$. What is ab ?

A. -17
B. 9
C. 12
D. 60

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Alice needs to replace a light bulb located 10 centimeters below the ceiling in her kitchen. The ceiling is 2.4 meters above the floor. Alice is 1.5 meters tall and can reach 46 centimeters above the top of her head. Standing on a stool, she can just reach the light bulb. What is the height of the stool, in centimeters?

A. 32
B. 34
C. 36
D. 38

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Of the 500 balls in a large bag, 80% are red and the rest are blue. How many of the red balls must be removed so that 75% of the remaining balls are red?

A. 25
B. 50
C. 75
D. 100

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Two non-zero real numbers, a and b, satisfy $ab = a - b$. Which of the following is a possible value of $\frac{a}{b} + \frac{b}{a} - ab$?

A. -2
B. $\frac{1}{3}$
C. $\frac{1}{2}$
D. 2

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

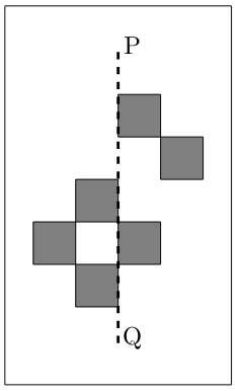
If all alligators are ferocious creatures and some creepy crawlers are alligators, which statement(s) must be true?

I. All alligators are creepy crawlers.
II. Some ferocious creatures are creepy crawlers.
III. Some alligators are not creepy crawlers.

A. I only
B. II only
C. III only
D. II and III only

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude



The least number of squares that must be added so that the line $P - Q$ becomes the line of symmetry is _____

A. 4
B. 3
C. 6
D. 7

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Points A and B are 10 units apart. Points B and C are 4 units apart. Points C and D are 3 units apart. If A and D are as close as possible, then the number of units between them is

A. 0
B. 3
C. 9
D. 11

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

Choose the most appropriate pair of words from the options given below to complete the following sentence.

She could not _____ the thought of _____ the election to her bitter rival.

A. Bear, loosing
B. Bare, loosing
C. Bear, losing
D. Bare, losing

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider the following two systems of linear equations :

S1 :
$x_1 + x_2 = 10$
$-x_1 + x_2 = 0$

S2 :
$x_1 - 2x_2 = -3$
$2x_1 - 4x_2 = 8$
Which of the above system has infinite solutions?

A. Only S1
B. Only S2
C. Both
D. None

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

If the given matrices $\mathrm{A} = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$ and $\mathrm{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ satisfy $\mathrm{A}^2 = k\mathrm{A} - 2\mathrm{I}$, the value of coefficient k is

_____

A. 1
B. 2
C. 0
D. 4

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Others

Let $\vec{u}_1$ be the first column and let $\vec{u}_2$ be the second column of the matrix U given from the SVD. Which of the following combinations of values of $\vec{u}_1$ and $\vec{u}_2$ are not valid?

A. $\vec{u}_1 = [1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]^T, \vec{u}_2 = [0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0]^T$
B. $\vec{u}_1 = [1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]^T, \vec{u}_2 = [0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0]^T$
C. $\vec{u}_1 = \left[\frac{1}{2} \quad \frac{1}{2} \quad 0 \quad 0 \quad 0 \quad 0\right]^T, \vec{u}_2 = \left[\frac{1}{2} \quad -\frac{1}{2} \quad 0 \quad 0 \quad 0 \quad 0\right]^T$
D. $\vec{u}_1 = [1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]^T, \vec{u}_2 = \left[\frac{1}{2} \quad \frac{1}{2} \quad 0 \quad 0 \quad 0 \quad 0\right]^T$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let E and F be two independent events, with $\mathrm{P}(\mathrm{E}) = \mathrm{P}(\mathrm{F}) = 0.25$. Let G be another event on the same sample space, which is disjoint from both E and F, and with $\mathrm{P}(\mathrm{G}) = 0.25$. Then, $\mathrm{P}(\mathrm{G}|\mathrm{F})$ is

A. 1
B. 0.5
C. 0.25
D. 0

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

You estimate the maximum possible outcome (number of dots) of rolling of a fair 6 -sided die using the outcome of a single roll of the die (pretending you don't already know the number of dots on the faces of the die).
What is the bias of this estimator?

A. 3.5
B. -2.5
C. 0
D. 2.5

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Let U and V be two independent zero mean Gaussian random variables of variances $1/4$ nd $1/9$ respectively. The probability ($3\mathrm{V} \geq 2\mathrm{U}$) is

A. $4/9$
B. $1/2$
C. $2/3$
D. $5/9$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Calculus

Find the value of the following limit?

$$\lim_{x \to 2} \frac{\sin\left(3x^2 - 12\right)}{x - 2}.$$

Your Answer: | Correct Answer: 12 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 1 | Penalty: 0 | Calculus

Evaluate the limit

$$\lim_{x \to \infty} (\sqrt{x+1} - \sqrt{x+2})$$

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

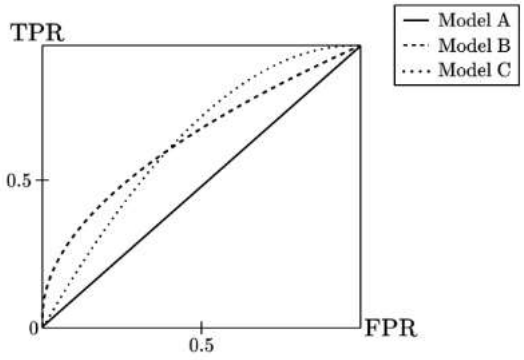
Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

Consider a function whose derivative is given by $f'(x) = x(x-1)$.
Which of the following option is/are true about the function?

A. There is a local max at $x = 0$
B. There is a local min at $x = 0$
C. There is a local $\max$ at $x = 1$
D. There is a local $\min$ at $x = 1$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning



Suppose we fix the decision threshold for all 3 logistic regression models to be such that we get a FPR of 0.8 when we evaluate our model. In order of most to least preferred, rank each of the models given their ROC curves.

A. $C > B > A$
B. $C > A > B$
C. $B > C > A$
D. $B > A > C$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 1 | Penalty: 0 | Others

Which of the following are considered linear models with respect to θ ?
Select all that apply.

A. $\hat{y} = \theta_1 x_1 + \theta_2 x_2$
B. $\hat{y} = \theta_0 + \theta_1 x_1^2$
C. $\hat{y} = \theta_1 + \theta_1^2 x_1$
D. $\hat{y} = \theta_0 + \theta_0 \theta_1 x_1$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Suppose we have a dataset $\{(x_i, y_i)\}_{i=1}^n$ and we model some particular observation y_i as:

$$\hat{y}_i = f_{\theta_1}(x_i) = \theta_1 \ln(x_i)$$

We are asked to find the minimizer $\hat{\theta}_1$ that minimizes the Mean Squared Error (MSE), defined as:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - f_{\theta_1}(x_i)\right)^2$$

What is the value of $\hat{\theta}_1$ that minimizes the MSE?

A. $\frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}$

B. $\frac{\sum_{i=1}^{n} \ln(x_i) y_i}{\sum_{i=1}^{n} \ln(x_i^2)}$

C. $\frac{\sum_{i=1}^{n} \ln(x_i) y_i}{\sum_{i=1}^{n} (\ln(x_i))^2}$

D. $\frac{\sum_{i=1}^{n} y_i / x_i}{\sum_{i=1}^{n} x_i^2}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Suppose we have a training dataset with $n = 2^6$ observations, consisting of some design matrix $\mathbb{X}$ and binary response variable y, and we want to train a binary classifier.
The all-zero classifier is a classifer that predicts 0 for all observations, regardless of input. The training accuracy of the all-zero classifier on our training data is $\frac{1}{8}$.
If we were to build a decision tree for classification, what would be the entropy of the tree at the root node, where all observations begin?

A. $-\frac{1}{8}\left[7\log_2\frac{1}{8} + \log_2\frac{7}{8}\right]$

B. $-\frac{1}{8}\left[\log_2\frac{1}{8} + \log_2\frac{7}{8}\right]$

C. $-\frac{1}{64}\left[8\log_2\frac{1}{8} + 7\log_2\frac{7}{8}\right]$

D. $-\frac{1}{8}\left[\log_2\frac{1}{8} + 7\log_2\frac{7}{8}\right]$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Suppose we have one qualitative variable that that we convert to numerical values using onehot encoding. We've shown the first four rows of the resulting design matrix below:

a	b	c
1	0	0
1	0	0

a	b	c
0	0	1
0	1	0

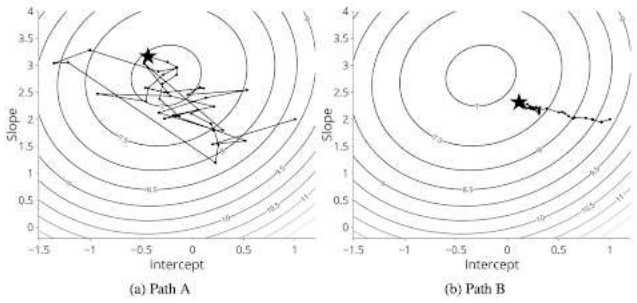
Say we train a linear model m_1 on these data. Then, we replace all of the 1 values in column a with 3 's and all of the 1 values in column $\mathbf{b}$ with 2 's and train a new linear model m_2. Neither m_1 nor m_2 have an intercept term. On the training data, the average squared loss for m_1 will be_____that of m_2.

A. greater than
B. less than
C. equal to
D. impossible to tell

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Exam Queries

The following figures depict the contours of the loss surface as well as the solution path of parameters obtained by running an optimisation algorithm. The star corresponds to the final parameter value. Use the following figures to answer the question.



(a) Path A (b) Path B

A. The sequence of parameters in Path A more likely came from an execution of Batch Gradient Descent, and Path B is Stochastic Gradient Descent.
B. Path A is Stochastic Gradient Descent, and Path B is Batch Gradient Descent.
C. Both Path A and Path B represent Batch Gradient Descent.
D. Both Path A and Path B represent Stochastic Gradient Descent.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Select the best description of the bias of an estimator $\hat{\theta}$ for a parameter θ, where $\hat{\theta}$ is computed based on a simple random sample from the population of interest.

A. The difference between θ and the value of $\hat{\theta}$ for a particular sample.
B. The average difference between θ and $\hat{\theta}$ computed for each element of the sample.
C. The average difference between θ and $\hat{\theta}$ over all possible simple random samples from the population.
D. The difference between the expected value of θ and the value of $\hat{\theta}$ for a particular sample.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

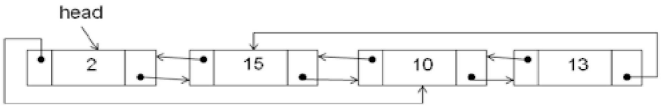
A recursive algorithm for a problem that reduces the problem on inputs of size n to 2 sub-problems of the same type of size $\frac{n}{3}$ plus $\Theta\left(n^2\right)$ extra work has running time:

A. $\Theta\left(n^2\right)$
B. $\Theta\left(n^2 \log n\right)$
C. $\Theta\left(n^{\log_2 3}\right)$
D. $\Theta\left(n^{\log_3 2}\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following linked list which has nodes of a doubly linked list type.



If we execute the following lines of code on a given linked list then what will be the output?

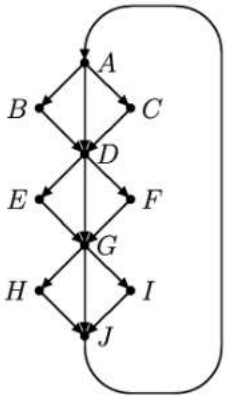
```
head.prev.next.next = head.next.next
head = head.prev.prev
head.prev = head
print(head.data)
```

A. 2
B. 15
C. 10
D. 13

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following directed graph.



List the nodes in the order they would be visited in a depth-first search of the graph starting at A. When choosing a node to explore next, break ties in favor of the alphabetically least.

A. ABCDEFGHIJ
B. ABDEGHJIFC
C. ABDEGHJFIC
D. ABDEGHJCFI

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 1 | Penalty: 0 | Others

Running DFS on a directed acyclic graph may produce:

A. Tree edges
B. Back edges
C. Forward edges
D. Cross edges

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Consider the following Python function where m and n are assumed to be a positive integers:

```
    def mystery(n,m):
        p = 0
        e = 0
        while e < m:
5.          p = p + n
            e = e + 1
        return p
```

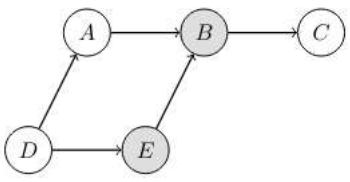
Which of the following functions is being computed by mystery above?

A. nm
B. $n + m$
C. n^m
D. m^n

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Consider the Bayesian network shown below. We want to infer $\mathbf{P}(A \mid b, e)$ where A is the query variable, B and E are evidence variables, and C and D are hidden variables. Which of the following statements is true?



A. $\mathbf{P}(A \mid b, e) \propto \sum_c \sum_d P(c \mid b)\mathbf{P}(b \mid A)P(b \mid e)\mathbf{P}(A \mid d)P(e \mid d)P(d)$
B. $\mathbf{P}(A \mid b, e) \propto \sum_c \sum_d P(c \mid b)\mathbf{P}(b \mid A, e)\mathbf{P}(A \mid d)P(e \mid d)$
C. $\mathbf{P}(A \mid b, e) \propto \sum_c \sum_d P(c \mid b)\mathbf{P}(b \mid A, e)\mathbf{P}(A \mid d)P(e \mid d)P(d)$
D. $\mathbf{P}(A \mid b, e) \propto \sum_b \sum_e P(c \mid b)\mathbf{P}(A \mid b, e)\mathbf{P}(A \mid d)P(e \mid d)P(d)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Consider the following two statements about the game search :
Statement 1 : An agent's environment is said to be stochastic if the next state is completely determined by the current state and the agent's action.
Statement 2 : In a zero-sum two-player game there is necessarily always a winner and a loser.

A. Both Statement 1 and Statement 2 are TRUE.
B. Only Statement 1 is TRUE
C. Only Statement 2 is TRUE
D. Both state 1 and statement 2 are FALSE

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Numerical Type | Award: 1 | Penalty: 0 | Artificial Intelligence

One approach to handling fuzzy logic data might be to design a computer using ternary $(\text{base-}3)$ logic so that data could be stored as "true," "false," and "unknown." If each ternary logic element is called a flit, minimum how many flits are required to represent at least 256 different values?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider the following instances:

QB1	
Q	Ans
1	x
2	y
3	y

QB2	
Q	Ans
1	x
1	y
2	y

Which of the following relational operations will produce the following tuple only?

3	y	1	y

A. $(\text{QB1} \cup \text{QB2}) \times (\text{QB2} \cup \text{QB1})$
B. $(QB1 - QB2) \cap (QB2 - QB1)$
C. $(\text{QB1} - \text{QB2}) \times (\text{QB2} - \text{QB1})$
D. $(QB1 \cap QB2) - (QB2 \cap QB1)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

Suppose B be the 10×10 matrix such that all entries are equal to 1. Then B has 2 eigenvalues : 10 (multiplicity 1) and 0 (multiplicity 9). Let $\mathrm{A} = (a_{ij})$ be a 10×10 matrix such that $a_{ij} = 1$ for $i \neq j$ and $a_{ii} = \alpha + 1$, where $\alpha > 0$. Let λ and μ be the largest and the smallest eigenvalues of A, respectively. If $\lambda + \mu = 24$, then α equals _____. Hint: Take the help of given information about eigenvalues of B.

Your Answer: | Correct Answer: 7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A be an $n \times n$ matrix with characteristic polynomial

$$p(\lambda) = \lambda(\lambda - 2)(\lambda - 3)^2.$$

Which of the following can you determine from this information?

A. The number n.
B. The trace of A.
C. The determinant of A.
D. The rank of A.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

A given matrix A is 5×5 with two distict eigenvalues.

$$\begin{bmatrix} \lambda_1 & 0 & 0 & 0 & 0 \\ 0 & \lambda_1 & 0 & 0 & 0 \\ 0 & 0 & \lambda_1 & 0 & 0 \\ 0 & 0 & 0 & \lambda_2 & 0 \\ 0 & 0 & 0 & 0 & \lambda_2 \end{bmatrix}$$

The number of linearly independent eigenvectors of A could be -

A. 1
B. 3
C. 5
D. ∞

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $\mathrm{A}x = b$ be a system of m linear equations in n unknowns, and let $r = \mathrm{rank}(\mathrm{A})$ and $s = \mathrm{rank}(\mathrm{A} \mid b)$.

Suppose that this system has a unique solution.

What can you say about the relative magnitudes of m, n, r, s?

A. $m \geq n$
B. $r = s$
C. $m > s$
D. $m > r - 1$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Determine all values of k such that

$$\left\{ \begin{bmatrix} 0 \\ -1 \\ k \end{bmatrix}, \begin{bmatrix} k \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

is a basis for $\mathbb{R}^3$.

A. $k \neq 0$
B. $k \neq 0, k \neq 2$
C. $k \neq 0, k \neq -2$
D. $k = 0, k = 2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ given in the standard basis by the matrix:

$$\begin{pmatrix} 2 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

Find a basis $\beta = \{v_1, v_2, v_3\}$ of $\mathbb{R}^3$ such that, with respect to the basis $\beta = \{v_1, v_2, v_3\}$ of $\mathbb{R}^3$ and the standard basis $\gamma = \{(1, 0), (0, 1)\}$ of $\mathbb{R}^2$, the matrix of T takes the form:

$$[T]_\beta^\gamma = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Note: e_i denotes the i-th column of the identity matrix in $\mathbb{R}^3$.

A. $v_1 = e_2, v_2 = e_3, v_3 = e_1 - 2e_2 + e_3$
B. $v_1 = e_2, v_2 = e_3, v_3 = e_1 + 2e_3$
C. $v_1 = e_1, v_2 = e_2, v_3 = e_1 - e_3$
D. $v_1 = e_1, v_2 = e_2, v_3 = e_2 - e_3$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $\mathrm{X}_i, 1 \leq i \leq 4$ be independent Bernoulli random variable each with mean $p = 0.1$.
Let $\mathrm{X} = \sum_{i=1}^{4} \mathrm{X}_i$. That is, X is a Binomial random variable with parameters $n = 4$ and $p = 0.1$. Then,

A. $P(\mathrm{X}_1 = 1 \mid \mathrm{X} = 2) = 0.1$
B. $P(\mathrm{X}_1 = 1 \mid \mathrm{X} = 2) = 0.5$
C. $P(\mathrm{X}_1 = 1 \mid \mathrm{X} = 2) = 0.25$
D. $P(\mathrm{X}_1 = 1 \mid \mathrm{X} = 2) = 0.6$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Let $\mathrm{P}(\mathrm{A}, \mathrm{B}) = \mathrm{P}(\mathrm{A} \cap \mathrm{B})$ and $\mathrm{P}\left(\mathrm{B}'\right) = \mathrm{P}\left(\mathrm{B}^c\right)$.
Which of the following formulas are ALWAYS true (without any condition on events A, B or C)?

A. $\mathrm{P}\left(\mathrm{A} \mid \mathrm{B}'\right) = 1 - \mathrm{P}(\mathrm{A} \mid \mathrm{B})$
B. $\mathrm{P}(\mathrm{A}, \mathrm{B} \mid \mathrm{C}) = \frac{\mathrm{P}(\mathrm{A}, \mathrm{B}, \mathrm{C})}{\mathrm{P}(\mathrm{C})}$
C. $\mathrm{P}(\mathrm{A} \mid \mathrm{B}, \mathrm{C}) = \mathrm{P}(\mathrm{A}, \mathrm{B} \mid \mathrm{B}, \mathrm{C})$
D. $\mathrm{P}(\mathrm{A}, \mathrm{B}, \mathrm{C}) = \mathrm{P}(\mathrm{A}|\mathrm{B}, \mathrm{C})\ \mathrm{P}(\mathrm{B}|\mathrm{C})\ \mathrm{P}(\mathrm{C})$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Consider the following Random Walk of an element. Suppose $y(0) = 0$; that is, $y = 0$ at time $t = 0$. The position of $y(n)$ depends probabilistically on its time at time $n - 1; y(n) = y(n-1) + 1$ with probability $1/2$, and $y(n) = y(n-1) - 1$ with probability $1/2$.

Which of the following is/are true?

A. Probability that $y(19) = 0$, is non-zero.
B. Probability that $y(n) = 0$, is 0 when n is odd.
C. Probability that $y(10) = 0$, is $\binom{10}{5} / 2^{10}$.
D. Probability that $y(2) = 0$, is 0.5.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Let E and F be the events that a family of n children has children of both sexes and has at most one boy, respectively. For which of the following value of n, E and F are Not independent?

A. 2
B. 3
C. 4
D. 5

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let x and y be random variables, not necessarily independent, that take real values in the interval $[0, 1]$. Let $z = xy$ and let the mean values of x, y, z be $\bar{x}, \bar{y}, \bar{z}$, respectively. Which one of the following statements is TRUE?

A. $\bar{z} = \bar{x}\bar{y}$
B. $\bar{z} \leq \bar{x}\bar{y}$
C. $\bar{z} \geq \bar{x}\bar{y}$

D. $\bar{z} \leq \bar{x}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $f(x) = x^{100} + x^{99} + \ldots x + 1$, then $f'(1)$ is equal to

A. 5050
B. 5049
C. 5051
D. 50051

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Professor Smith has a dataset with one feature, shown in the table below:

1	1	1	3	4	5	6	8

Professor Smith wants to try a new clustering method called k-Medians, which works the same as k-Means, but with the cluster centers being located at the median of their data points instead of the mean. They initialize 2 clusters centered at 2 and 7 .
After assigning each point to one of these clusters (i.e., after running only first iteration), what are the values of points located in the cluster centered at 7 ?

A. 6,8
B. $5, 6, 8$
C. $4, 5, 6, 8$
D. 5,6

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Consider logistic regression being applied to data with n points for binary class 0 and 1 . Let y_i be the ground truth, and $\hat{y}_i$ be the probability that x_i belongs to class 1 .
Suppose, instead of using cross-entropy, we decide to use the following loss function:

$$L(\theta) = \frac{1}{n} \sum_{i=1}^{n} y_i \left(y_i - \hat{y}_i\right) + \left(1 - y_i\right) \left(y_i - \hat{y}_i\right)$$

Which of the following is/are true about this loss function? Select all that apply.

A. Under this loss function, worse model predictions can incur less penalty than better model predictions.
B. A model that always predicts $\hat{y} = 0$ will always have a loss ≥ 0.
C. The range of this loss function is $[-1, 1]$.
D. A false positive will incur a loss of -1 , and a false negative will incur a loss of 1 .

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Given the set of points $\{-15, 10, 20, 30, 30, 35, 40, 50\}$, and the loss function $\sum |x_i - c|$, what is the optimal value of c that minimizes the corresponding loss?

A. 25
B. 50
C. 40
D. 30

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Assume you have two positive increasing functions f and g such that $f(n)$ is in $O(g(n))$. For each of the following statements, decide which one(s) is/are always TRUE.

A. $2^{f(n)}$ is $O\left(2^{g(n)}\right)$
B. $f(n)^2$ is $O\left(g(n)^2\right)$
C. $f(n) = O\left((f(n))^2\right)$
D. $g(n) = \Omega(g(n))$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Numerical Type | Award: 2 | Penalty: 0 | Others

How many ways are there to insert the five elements $1, 2, 3, 4, 5$ into a binary search tree such that the resulting tree has a height of 4 ? The height is defined as the number of edges between the root and the farthest leaf.

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Given a circular array-based queue capable of holding 100 objects. Suppose the queue is initially empty, and then objects are put into the queue at the rate of 10 per minute while meantime they are processed and removed from the queue at the rate of 5 per minute. After 120 elements have been added to the queue, which of the following is true?

A. You can't add 120 elements to an array holding 100 entries.
B. There will be 60 elements in the queue, 20 of them at the front of the array where the queue started, and 40 at the other end.
C. There will be 60 elements in the queue, 30 of them at the front of the array where the queue started, and 30 at the other end.
D. There will be 60 elements in the queue, 40 of them at the front of the array where the queue started, and 20 at the other end.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider the following list of elements:

$$8 \quad 1 \quad 5 \quad 2 \quad 6 \quad 3 \quad 7 \quad 4$$

What will be the total number of comparisons between the list elements required to sort this list using the Merge Sort algorithm?

A. 12 comparisons
B. 16 comparisons
C. 17 comparisons
D. 20 comparisons

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

In a breadth-first search (BFS) of a directed graph, where $v.d$ represents the distance from the source to v. Which of the following is/are true?

A. There are no forward edges.
B. For each tree edge (u, v), we have $v.d = u.d + 1$.
C. For each tree edge (u, v), we have $v.d \leq u.d + 1$.
D. For each tree edge (u, v), we have $v.d \leq u.d$.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Suppose that only numbers in $1 \ldots 1000$ appear as keys in a binary search tree. While searching for 500 , which of the following sequences of keys could not be examined?

A. $300, 400, 900, 800, 500$
B. $700, 200, 600, 550, 500$
C. $200, 700, 600, 300, 400, 500$
D. $100, 1000, 200, 800, 300, 900, 500$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Numerical Type | Award: 2 | Penalty: 0 | Programming in Python

Consider the following Python function:

```
def mystery(t):
    if len(t) == 0:
        return 0
    else:
5.      return (((t[0] % 2) + 1) % 2) + mystery(t[1:])
```

What will be the output of the function mystery([2, 5, 7, 6, 4])?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

Consider the following Python code:

```
l = 'it was the best of times it was the worst of times'.split()

d = {x[0]: [] for x in l}
for y in l:
    d[y[0]].append(y)
```

What does d['w'] evaluate to, after executing the above code?

A. []
B. ['worst']
C. ['was', 'was', 'worst']
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose X is a random variable with $P(X = k) = (1 - p)^k p$ for $k \in \{0, 1, 2, \ldots\}$ and some $p \in (0, 1)$. For the hypothesis testing problem

$$H_0 : p = \frac{1}{2} H_1 : p \neq \frac{1}{2}$$

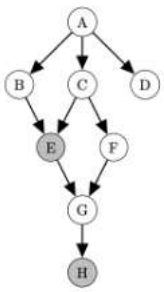
consider the test "Reject H_0 if $X \leq A$ or if $X \geq B$ ", where $A < B$ are given positive integers. The type-I error of this test is

A. $1 + 2^{-B} - 2^{-A}$
B. $1 - 2^{-B} + 2^{-A}$
C. $1 + 2^{-B} - 2^{-A-1}$
D. $1 - 2^{-B} + 2^{-A-1}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Assume we are given the following Bayes' net, and would like to perform inference to obtain $P(B, D \mid E = e, H = h)$.



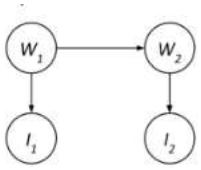
What is the number of rows in the largest factor generated by inference by enumeration, for this query $P(B, D \mid E = e, H = h)$?
Assume all the variables are binary.

A. 2^2
B. 2^3
C. 2^6
D. 2^8

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Numerical Type | Award: 2 | Penalty: 0 | Artificial Intelligence

We would like to analyze people's ice cream eating habits on sunny and rainy days Suppose we consider the weather, along with a person's ice-cream eating, over the span of two days. We'll have four random variables: W_1 and W_2 stand for the weather on days 1 and 2 , which can either be rainy R or sunny S , and the variables I_1 and I_2 represent whether or not the person ate ice cream on days 1 and 2, and take values T (for truly eating ice cream) or F. We can model this as the following Bayes Net with these probabilities.



W_1	$P(W_1)$
S	0.6
R	0.4

W_1	W_2	$P(W_2 \mid W_1)$
S	S	0.7
S	R	0.3
R	S	0.5
R	R	0.5

W	I	$P(I \mid W)$
S	T	0.9
S	F	0.1
R	T	0.2
R	F	0.8

Suppose we produce the following samples of (W_1, I_1, W_2, I_2) from the ice-cream model:

$R,\ \mathrm{F}, \mathrm{R}, \mathrm{F}$ $\quad \mathrm{R}, \mathrm{F}, \mathrm{R}, \mathrm{F}$ $\quad \mathrm{S}, \mathrm{F}, \mathrm{S}, \mathrm{T}$ $\quad \mathrm{S}, \mathrm{T}, \mathrm{S}, \mathrm{T}$ $\quad \mathrm{S}, \mathrm{T}, \mathrm{R}, \mathrm{F}$
$\mathrm{R}, \mathrm{F}, \mathrm{R}, \mathrm{T}$ $\quad \mathrm{S}, \mathrm{T}, \mathrm{S}, \mathrm{T}$ $\quad \mathrm{S}, \mathrm{T}, \mathrm{S}, \mathrm{T}$ $\quad \mathrm{S}, \mathrm{T}, \mathrm{R}, \mathrm{F}$ $\quad \mathrm{R}, \mathrm{F}, \mathrm{S}, \mathrm{T}$

What is $\widehat{P}\,(W_2 = \mathrm{R})$, the probability that sampling assigns to the event $W_2 = \mathrm{R}$? (upto 2 decimal places)

Your Answer: | Correct Answer: 0.50 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Let $P(x), Q(x), R(x)$ and $S(x)$ denote the following predicates with domain $\mathbb{Z}$:

$$P(x) : x \leq 0,$$
$$Q(x) : x^2 = 1,$$
$$R(x) : x \text{ is odd},$$
$$S(x) : x = x + 1.$$

Which of the following statements is/are true?

A. $\forall x \in \mathbb{Z}, \quad R(x) \rightarrow S(x)$
B. $\forall x \in \mathbb{Z}, \quad S(x) \rightarrow R(x)$
C. $\exists x \in \mathbb{Z}$ such that $Q(x) \wedge \sim R(x)$
D. $\exists x \in \mathbb{Z}$ such that $P(x) \rightarrow S(x)$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

An attribute A is called prime if A is in any of the candidate keys. Assume in the following statements, A is an attribute & $\mathcal{X}$ is a set of attributes.

Consider the following statements:

A. Domain of each attribute is an elementary type; that is, not a set or a record structure.

B. Whenever $\mathcal{X} \mapsto A$ is a functional dependency that holds in relation $\mathbf{R}$ and $A \notin \mathcal{X}$, then either
- A is prime, or
- $\mathcal{X}$ is a key or a super-key for $\mathbf{R}$.

C. Whenever $\mathcal{X} \mapsto A$ is a functional dependency that holds in relation $\mathbf{R}$ and $A \notin \mathcal{X}$, then either
- A is prime, or
- $\mathcal{X}$ is not a proper subset of any key for $\mathbf{R}$.

D. Whenever $\mathcal{X} \mapsto A$ is a functional dependency that holds in relation $\mathbf{R}$ and $A \notin \mathcal{X}$, then
- $\mathcal{X}$ is a key or a super-key for $\mathbf{R}$.

Which of the following is a correct match??

A. $\mathrm{A-1NF, B-2NF, C-3NF, D-BCNF}$
B. $\mathrm{A-1NF, B-BCNF, C-2NF, D-3NF}$
C. $\mathrm{A-1NF, B-3NF, C-2NF, D-BCNF}$
D. $\mathrm{A-BCNF, B-3NF, C-2NF, D-1NF}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the following tables:

$$R = \begin{pmatrix} a & b \\ 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad S = \begin{pmatrix} a & b \\ 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad T = \begin{pmatrix} a & b \\ 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$\mathrm{SELECT}\ R.a, R.b, S.b, T.b\ \mathrm{FROM}\ R, S, T\ \mathrm{WHERE}\ R.b = S.a \quad \mathrm{AND} \quad S.b \neq T.b;$
Compute the results. Which of the following is/are true: A. $(0,1,1,0)$ appears twice. B. $(1,1,0,1)$ does not appear. C. $(1,1,1,0)$ appears once. D. $(1,1,1,1)$ does not appear.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Numerical Type | Award: 2 | Penalty: 0 | Databases

A B-Tree of order m is an m-way search tree with the following properties:

- The non-leaf nodes store up to $m-1$ keys to guide the searching.
- The root is either a leaf or has between 2 and m children.
- All non-leaf nodes (except the root) have between $\lceil m/2 \rceil$ and m children.

The record pointer, key field and block pointer sizes of a B-Tree are 8 bytes, 10 bytes, and 6 bytes, respectively. If the block size is 1 KB , then the order of the tree is ___

Your Answer: | Correct Answer: 43 | Not Attempted | Time taken: 00min 00sec | Discuss

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